Unit 1

1. What is data science? Explain the data analytics pipeline with examples.
2. What is knowledge? Explain the process of KDD in data analysis.
3. Compare structured and unstructured data with examples.
4. Why is data processing important in data analysis? Explain the different types of data processing methods with examples.
5. Explain in detail how missing data are handled with examples.
6. Differentiate between standardization and normalization.
7. For the following sets of data, perform standardization and normalization.

Price (in Rs)	23	41	32	16	4

8. Explain why discretization is necessary?

Unit 2

1. Explain different types of data with examples. For the given data what analysis can be done. (Hint: Categorical and Numerical)
2. Explain power law distribution with real life examples. Why are power laws called heavy tailed distributions?
3. What is empirical distribution? Explain it for numerical and categorical data.
4. Explain Pearson Correlation and also describe how it describes the relationship between two continuous variables.
5. For the following dataset, calculate Pearson correlation coefficient between age and salary and interpret the result.

Age	**Salary (in Years)**
25	30000
32	45000
40	55000
48	65000
55	75000

6. What is statistical significance? Explain the use case of t-test, chi-squared test with examples.
7. Find the biserial correlation coefficient for the following data.

Study Hours	15	8	20	5	18	10	22
Result	Pass	Fail	Pass	Fail	Pass	Fail	Pass

8. The following table presents a cross tabulation between food preferences and gender.

	Male	Female	Other
Likes veg momo	8	75	1
Likes buff momo	63	3	0
Likes chicken momo	23	19	3

Do you think food preferences are correlated with gender? Justify using chi-squared test with significance level 0.05. The critical value for a chi-squared test with 4 degrees of freedom is 9.4877.

Unit 3

1. Compare simple and multi-variate linear regression with examples. Explain the significance of R^2 and Adjusted R^2.
2. What is Ordinary Least Square Estimation? How is multivariate linear regression evaluated using the matrix method?
3. What do you understand by multicollinearity? Explain how those multicollinearities are handled?
4. Fit the following data in a simple linear regression model.

Observation	1	2	3	4	5
X	2	4	5	7	9
Y	5	7	12	15	18

   a) Find the line of regression using the matrix method.
   b) Predict the value of Y when X = 8
5. For the following data, estimate the value of y at x = 8 using the Nadaraya-Watson estimator. Use h = 0.02

x	0.4	0.9	1.4	1.9
y	-1.8	1.2	8.5	19.1

6. What is PCA? Explain in detail how PCA reduces the dimension.
7. For the following dataset, use PCA to determine the principal components for the following data.

X	2	3	4	5	6	7
Y	1	5	3	6	7	8

Unit 4

1. What is a decision tree? Explain in detail about the cart algorithm.
2. For the following data, create a decision tree using both entropy (ID3) and gini (CART).

A	T	T	T	T	T	F	F	F	T	T
B	F	T	T	F	T	F	F	F	T	F
Class	Black	Black	Black	White	Black	White	White	White	White	White

3. For the following data, create a decision tree using both entropy (ID3) and gini (CART).

Age	young	young	middle	senior	senior	middle
Income	high	medium	high	low	medium	low
Purchased	no	yes	yes	no	yes	yes

4. Explain what is logistic regression? How does logistic regression classify the data?
5. Explain variable importance permutation test.
6. What is clustering? Explain how k-means clustering works?
7. For the following data, perform k-means clustering.

X	2	3	4	5	6	7
Y	1	5	3	6	7	8

Assume k=2, determine the clusters.

Unit 5

1. What is time series data? Explain the components of time series.
2. What is a stationary point? Explain the difference between ARIMA and ARMA.
3. What is granger causality? Explain how event A causes event B in time series data.
4. Compare and contrast between causality and correlation.
5. Explain ACF and PACF in terms of time series data.
6. What is AIC and BIC? How does it validate time series data?
7. Discuss the ARMA and ARIMA model and the use of stationary points.